# MFML - Mid Sem EC2
# Date: 20 Sep 2026

Question

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Consider the augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 2 & 1 & 4 \\ 2 & 4 & 2 & 8 \\ 1 & 1 & -1 & 0 \end{array}\right]$$

a) Convert it into row echelon form.

b) Determine the rank of the coefficient matrix.

c )Find all solutions.

**(3 Marks)**

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Consider the subspace

$$W = \{(x, y, z) \in \mathbb{R}^3 : x + 2y - 3z = 0\}$$

of $\mathbb{R}^3$. Show that

$$\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix} \right\}$$

is a basis of $W$.

**(3 Marks)**

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Three vectors representing directions in a vector space are given by

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \qquad v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \qquad v_3 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}.$$

Using the Gram–Schmidt process:

(a) Construct a set of mutually orthogonal vectors from $v_1, v_2, v_3$.

(b) Normalize the resulting vectors to obtain an orthonormal basis.

(c) Verify that the resulting vectors are mutually orthogonal by computing their pairwise inner products and verify that each vector in the orthonormal basis has unit length.

**(3 Marks)**

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Consider a matrix $\mathbf{A} \in \mathbb{R}^{3\times 3}$ defined as follows:

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Derive all the eigenvalues of **A** and the corresponding eigenvectors. Also, derive the dimension of the eigenspace corresponding to each eigenvalue of **A**. **(3 marks)**

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Consider the given data matrix $\mathbf{A} = \begin{bmatrix} 12 & 4 \\ 4 & 6 \end{bmatrix}$.

i) Compute the eigenvalues of the given data matrix. Identify the largest eigenvalue (in absolute value) and find an eigenvector of unit norm corresponding to it. **(2 Marks)**

ii) Let $\mathbf{v} \neq \mathbf{0}$ be an eigenvector of a square matrix **A** corresponding to the eigenvalue $\lambda$. Prove that **v** is also an eigenvector of $\mathbf{A}^2$ corresponding to eigenvalue $\lambda^2$. **(0.5 Mark)**

iii) Using results obtained in i) and ii), find first singular value, first left singular vector $\mathbf{u}_1$ and first right singular vector $\mathbf{v}_1$ of the given data matrix. **(1.5 Marks)**

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Consider the scalar field $f(x, y) = x^2 y^3$.

(a) Compute the gradient vector $\nabla f$ at the point (2,1). **(1 Mark)**

(b) Calculate the directional derivative of $f(x, y)$ at the point (2,1) in the direction of the vector $\mathbf{v} = 3\mathbf{i} + 4\mathbf{j}$. **(2 Marks)**

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Consider the function

$f(x) = (x^2 + 3x)^2 + \ln(x^2 + 3x)$.

Introduce the intermediate variables

$a = x^2,\ b = 3x,\ c = a + b,\ d = c^2,\ e = \ln c,\ f = d + e.$

Using reverse-mode automatic differentiation (backpropagation) on this computation graph:

(a) Draw the computation graph.
(b) Perform the forward pass at $x = 1$ and report the numerical value of every intermediate variable $a, b, c, d, e$ and of $f$.
(c) Write the local derivative of every node with respect to its parent(s) and evaluate them at $x = 1$.
(d) Perform the backward pass, i.e. compute $\frac{\partial f}{\partial d}, \frac{\partial f}{\partial e}, \frac{\partial f}{\partial c}, \frac{\partial f}{\partial a}, \frac{\partial f}{\partial b}$ and finally $\frac{df}{dx}$ at $x = 1$.

(Numerical values of ln 4 etc. may be left in symbolic form.)

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Answer the following with justifications.

a) Determine whether the set
$$V = \{(x, y) \in \mathbb{R}^2 : x \geq 0,\ y \geq 0\}$$
is a vector space over $\mathbb{R}$ under the usual addition and scalar multiplication giving adequate justification(s). (0.5 marks)

b) Choose the correct alternative and fill in the blank with justifications. The null space of a matrix $A_{m\times n}$ (is a subspace / is not a subspace) of _____. (0.5 marks)

c) Let $V = \text{span}\{(1, 2, 1), (2, 4, 2), (1, 1, 0), (3, 5, 2)\} \subseteq \mathbb{R}^3$. Find a basis for $V$ and determine $\dim(V)$. (2 marks)

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Prove or disprove the following.

a) The product of two lower triangular matrices of order $n \geq 2026$ is lower triangular.
b) We cannot construct a non-square matrix $A$ having minimum 3 rows and 4 columns such that $A^TA$ is positive definite.
c) The geometric multiplicity of an eigenvector is always more than it's algebraic multiplicity.
d) $LU$ decomposition can be performed on all square matrices.

**(4 Marks)**